AI-GENERATED RESEARCH MANUSCRIPT

Fixed-Format Undecidability of Exact Catalytic Tensor Restriction

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Selected research / Mathematical foundations

AI-generated mathematical proof candidate

Fixed-format undecidability of exact catalytic tensor restriction

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Abstract

An AI-generated proof candidate encoding finite-dimensional algebra representations into exact catalytic restriction of three-tensors. The proposed fixed positive rank-one-affine family is recursively enumerable complete. Mathematical correctness and historical priority remain open to independent scrutiny. The principal construction uses labelled rank-drop hyperplanes to prevent arbitrary local maps from evading a matrix-algebra representation constraint. We present its proof structure, strict scope, and the exact source for detailed checking.

Verification and authorship

This is an AI-generated research manuscript reporting a complete candidate argument . The original source contains mathematical proofs, but the universal construction has not passed an independent end-to-end specialist or formal proof check, and historical novelty has not been established. Source-reported small exact checks were not independently rerun. It should not be cited as a verified landmark theorem.

Harjas Sandhu developed and coordinates the agentic research system and is the contact person, not a scientific author of the AI-developed mathematics. No physical catalyst, laboratory implementation or practical quantum advantage is claimed.

1. Exact conversion contract

Over the complex numbers, write a three-party tensor as a matrix pencil T = SUMi ei tensor Ai . Tensor restriction T >= S allows one linear map on each party, without requiring those maps to be invertible. For a positive integer d define its bipartite maximally entangled amplification:

T[d] = ?? e? ? (A? ? I_d)
T ?cat S ? there exists d ? 1 such that T[d] ? S[d].

The auxiliary state is shared only between parties two and three and is returned exactly,

uncorrelated with the target, on the successful branch. The probability of success may be arbitrarily small. No approximate return, deterministic LOCC guarantee or three-party auxiliary state is assumed.

2. Main proposed theorems

Representation universality

From any finite rational presentation ?? = Q?X1,?,X??/(p1,?,ps) , the construction effectively produces integer tensors T(??) and S(??) for which, at every catalyst dimension d ,

T(??)[d] ? S(??)[d]
? ?? admits a unital representation in M_d(?).

The representation need not be faithful. Arithmetic-circuit relations can be converted to quadratic relations without changing which finite-dimensional representations exist.

Fixed positive rank-one affine undecidability

There exist fixed strictly positive integer tensors U, V and S, with U and V having a common format and tensor rank(V) = 1, such that the integer set

{ n ? 0 : U + nV ?cat S }

is recursively enumerable complete. Thus each finite-dimensional instance is algebraic and decidable, but permitting an unknown, arbitrarily large catalyst makes the overall exact-conversion decision undecidable.

3. Construction and proof

3.1 Matrix representations as two-sided equations

Quadraticise the presentation and clear rational denominators. With X0 = I, write each relation as a sum of terms c??X?X?. Introduce matrix units E?? and equations P(Mi tensor I_d)Q = biI_d, where P and Q consist of unknown d-by-d blocks. The normalization M0 = E00 forces P0Q0 = I. Additional E?0 - E0? relations identify the variables P0-1P? and Q?P0. Every algebra relation then becomes a two-sided equation, and the construction is reversible.

3.2 Rank-drop markers prevent slice mixing

The difficult step is that arbitrary restriction maps can mix pencil slices. For every index i other than a distinguished zero slice *, adjoin the linear forms x* + xi and 2x* + xi, together with x* itself. Give the successive marker blocks strictly increasing sizes h? = (r+1)j, where r is the number of core block variables.

D(x) = direct sum over j of [??(x) I_(h?)]
A(x) = core-source(x) ? D(x)
B(x) = core-target(x) ? D(x).

The target pencil is concise in its first factor. Any first-factor map G in a restriction must be invertible. On a marker hyperplane ker ??, the source rank falls by at least h? - r. Rank monotonicity forces G to permute these hyperplanes. Since adjacent block sizes differ

by r+1, the permutation can only be the identity. The labels ? = x*, x*+xi and 2x*+xi then force all dual eigenvalues to agree, so G is scalar. This is the key proposed new mechanism.

3.3 Saturating the marker block prevents a false representation

At the distinguished slice *, the core vanishes but every marker block is invertible. Its output has the full marker rank Kd, leaving no freedom for marker coordinates to contribute to the target core. Compressing all other slice equations to the output core therefore recovers exactly P(Mi tensor I_d)Q = biI_d. Conversely, any solution of these equations extends to a tensor restriction by identity maps on the marker blocks. This completes the representation compiler, provided the block-rigidity reasoning is correct.

3.4 Integrality and the halting reduction

Use the ordinary finite-dimensional representation theory of sl2 and a central generator z constrained by z? = (h+1)? + 4fe. On every simple finite-dimensional representation, z takes a nonzero integer scalar value, and every such integer is attainable. Differences of two such scalars encode arbitrary integer variables. The classical Davis-Putnam-Robinson-Matiyasevich theorem then provides one fixed polynomial P(n,x) encoding an recursively enumerable complete set.

Make n enter only through the relation t - nI = 0. After quadraticisation, the number of variables and relations is fixed; a single source coefficient is affine in n. The representation compiler consequently produces one fixed target and one rank-one source direction V.

3.5 Strictly positive coefficients

A fixed change of basis selects one corner contraction of both U and V equal to 1. For a fixed integer tensor W with corner coefficient 1 and all coefficients bounded in magnitude by M, apply to every factor the invertible map G = I + L?1?e0? with L = 8M + 1. Every transformed coefficient equals L? + ? with |?| <= M(3L?+3L+1) < L?. Thus U, V and S may be taken strictly positive without changing their conversion relation or tensor rank(V).

3.6 A concrete witness to genuine catalysis

The relations X? = Y? = I and XY + YX = 0 have no scalar complex solution, but do have a two-dimensional solution given by Pauli matrices. The claimed compiler gives a 7x367x367 source and a 7x365x365 target that are not convertible without a catalyst but are convertible with d=2. This is an exact algebraic witness, not an experimental success-rate claim.

The detailed line-by-line original derivation, including marker dimensions and coefficient calculations, is preserved in the complete AI source narrative . Readers evaluating the theorem should check the original equations rather than relying solely on this explanatory transcription.

4. Consequences and research frontiers

The proposed fixed-format undecidability excludes every computable bound on the necessary catalyst dimension as a function of input size. It also excludes a sound, effectively enumerable, complete certificate family for all impossible exact conversions. Restricting U+nV to real parameters gives a nonsemialgebraic conversion set; otherwise membership on sufficiently large integers would stabilise, contradicting the halting encoding.

The decisive research extension is to determine whether any such obstruction survives physically meaningful approximation, bounded success probability and chargeable catalyst preparation . The exact theorem by itself neither demonstrates a quantum computational speedup nor implies that practical converters require unbounded resources. An approximation theorem or an effective fixed-noise cutoff would be a fundamentally different result.

5. Comparison and limitations

Known stochastic multipartite entanglement catalysis is not new. This proof candidate should be compared with Chen et al. (2010) , Christandl, Gesmundo, Lysikov and Steffan , and Steffan’s 2023 thesis . The claimed contribution is the rank-drop representation compiler and its fixed-format decidability consequences, not the existence of quantum catalysis or the classical undecidability of Diophantine equations.

No result is claimed for approximate conversion, a dimension-independent lower bound on success probability, deterministic LOCC, a full three-party catalyst, or the matrix multiplication exponent. Correctness and historical originality require specialist assessment.

6. Research provenance and disclosure

The core construction comes from Astra result N680 (11 October 2026). The originating AI research session supplied the representation-compiler proof narrative, fixed-affine reduction, integer integrality gadget, and finite examples. This website transcription and its accompanying paper were AI-generated from that mathematical source under human research coordination. No independent human mathematical authorship, laboratory demonstration, journal acceptance, or external specialist verification is asserted.

Citation

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